

Basic Principles of Medicine 1

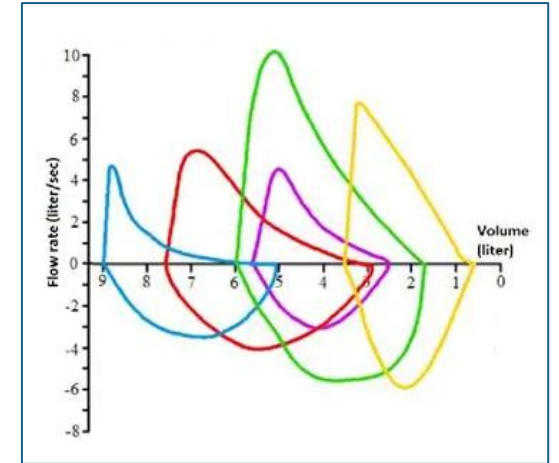
Module: CPR

Pulmonary DLA 15



Topic: Airway and Airflow

(notes to go with online video)



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Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0208. Describe the effects of airway diameter and turbulent flow on airway resistance.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0209. Explain dynamic compression of the airways.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0210. Use the concept of dynamic compression of airways to explain why each point in the effort- independent region of the curve represents a maximal flow rate that is uniquely dependent on lung volume.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0211. Describe how and why the shape of the flow-volume curve is shifted in chronic obstructive or restrictive lung disease (COPD).

SOM.MK.II.BPM1.3.CPR.2.PHYS.0212. Explain how mechanical changes with restrictive and obstructive disease affect the EPP.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0346. Explain the equal pressure point (EPP) and the forces that contribute to its formation during a forced expiration.

Assigned Readings

Medical Physiology: Principles for Clinical Medicine. Sixth Edition. Rodney A. Rhoades, David R. Bell. Chapter 18: Ventilation and the Mechanics of Breathing. Read under the heading: Airway resistance and Work of Breathing.

Medical Physiology, Third Edition. Boron, Walter F., MD, PhD. Chapter 27: Mechanics of Ventilation. Read under the heading: Dynamic Properties of the Lung.

Video link:

<https://sgu1.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=a5473b55-b558-472f-bdab-b35c00cd9e02> *(please copy and paste this link in browser; clicking on this may not work!)*

Airway Resistance

Definition: Airway resistance is the major factor opposing the air flow in and out of the lungs

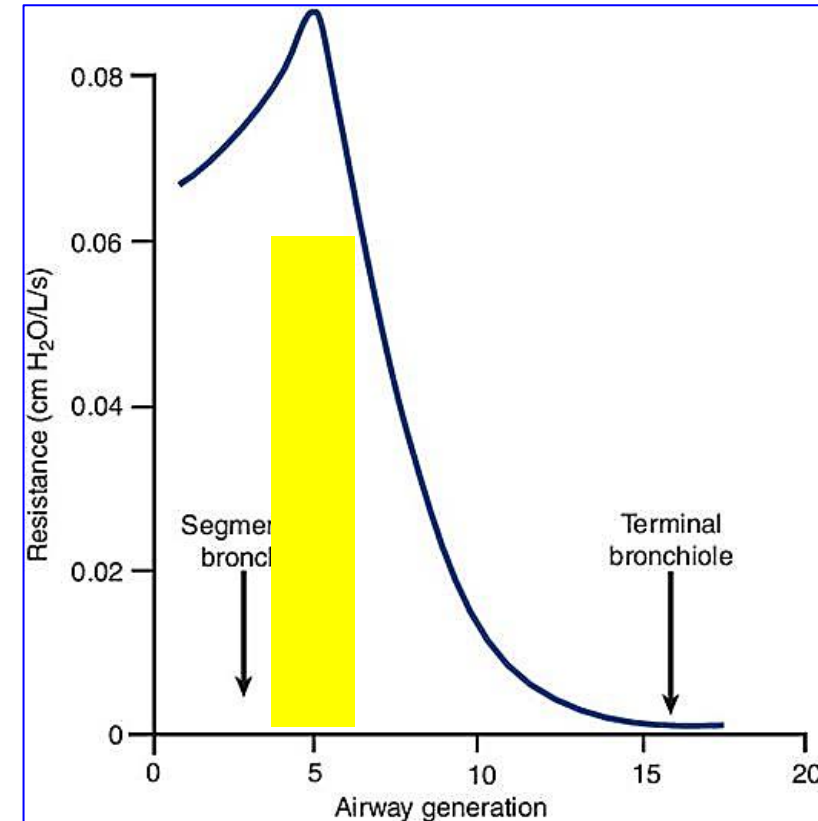
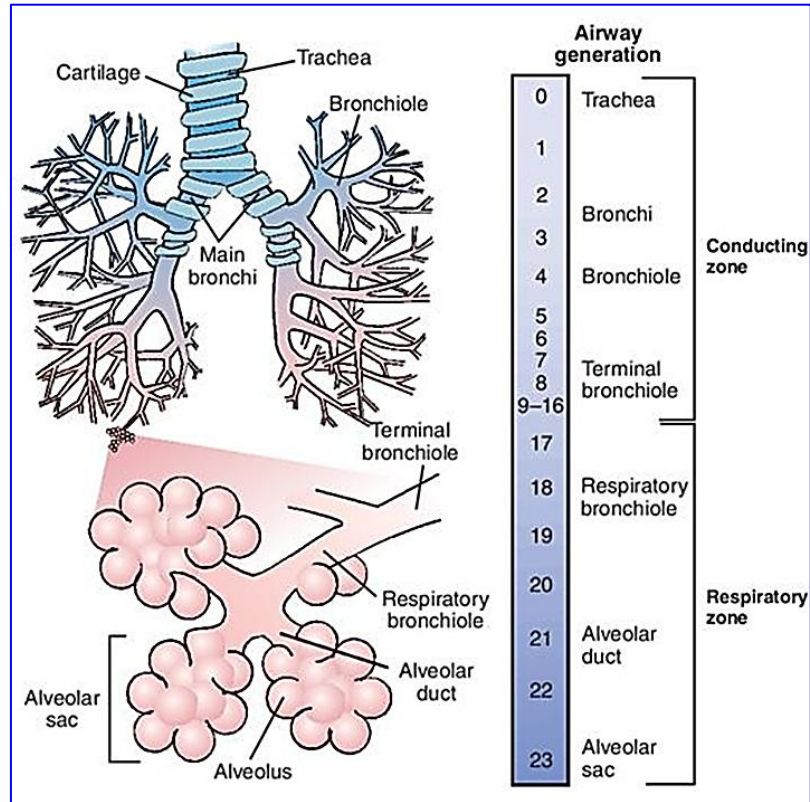
$$\text{Resistance} = \text{Pressure gradient} / \text{Flow}$$

The pressure gradient between alveoli and mouth is the driving force for the flow.

Laminar and Turbulent Flow

- In laminar flow, resistance is inversely proportional to the fourth power of the radius of the tube.
- In laminar flow, the velocity profile shows a central spike of fast movement.
- Turbulent flow is most likely to occur at high Reynolds numbers.

The major sites of airway resistance for decreasing airflow are the bronchi



- The major sites of resistance in the airways are from the **medium-sized bronchi** (lobar and segmental bronchi) down to about 7th generation (upper terminal bronchioles).
- **Turbulence** in these segments significantly increases airway resistance.
- **Small airways** (mostly terminal and respiratory bronchioles) offer least airway resistance.

Small airways offer least resistance-Why

A single small airway (bronchioles) may contribute a large resistance due to small cross-sectional area 

However, in lungs:

- Many small airways put together greatly **increase total cross-sectional area**, and
- Small airways are **arranged parallel to each other** so that the total resistance (R_T) is sum of the reciprocals of individual resistances



$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Above 2 factors make the bronchioles to offer least resistance.

Factors affecting Airway Resistance

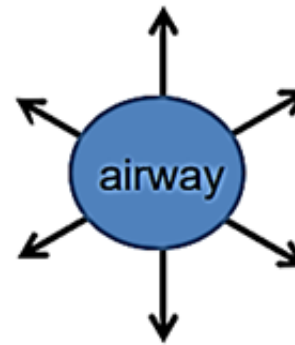
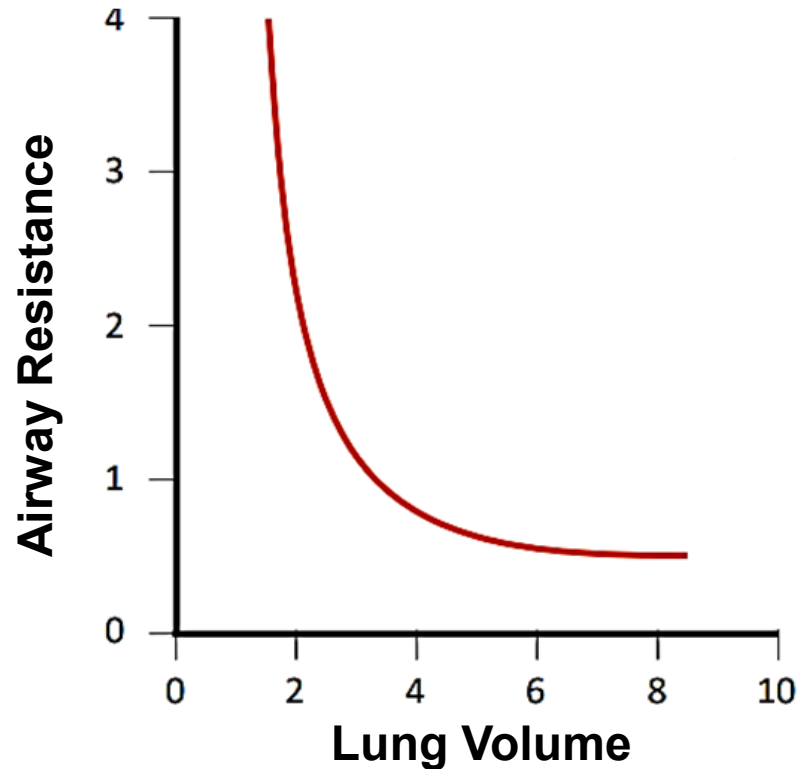
1. Lung volume
2. Bronchial smooth muscle tone
3. Dynamic compression by changes in pleural and alveolar pressures

Mechanism:

- First 2 factors alter the radius of the small airways.
- Change in **radius of the small airways has the greatest effect** on the resistance.
- Dynamic compression (third factor in the list) offers resistance mostly in diseased lungs (see details later) not in healthy lungs.

Relationship between Lung Volume and Airway Resistance

Change in radius has the greatest effect on the resistance

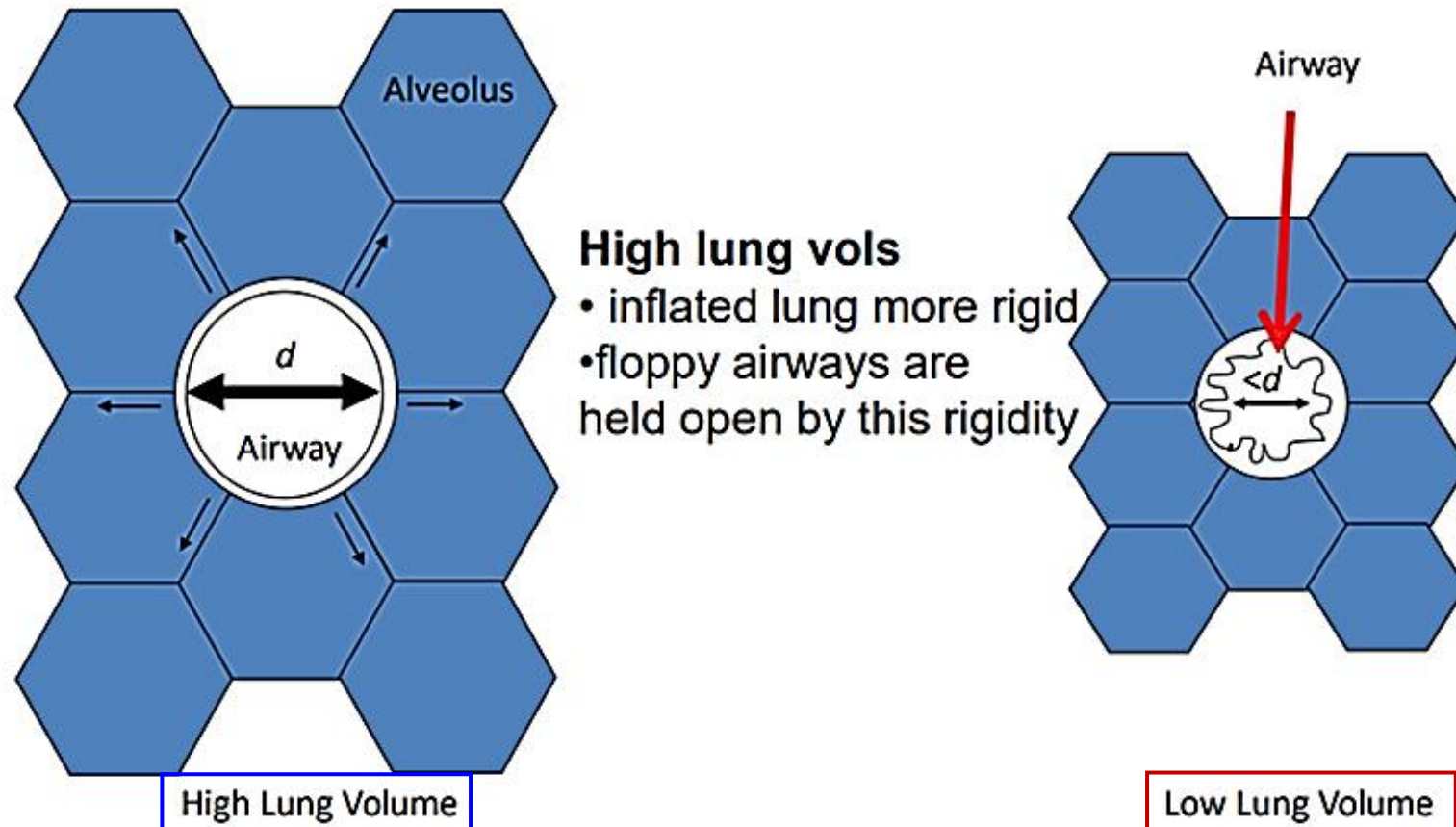


Traction on small airways by elastic recoil of distending alveoli increase the diameter of these airways. This is called **radial traction**.

- At **high lung volumes**, there is **increased radial traction** on small airways pulling them open and therefore the airway resistance decreases.
- At **low lung volumes**, **decrease in distending force (radial traction)** on airways reduce the radius of the small airways and thus increase airway resistance.

Relationship between Lung Volume and Airway Resistance

Change in radius has the greatest effect on the resistance



Patients with increased airway resistance frequently have high lung volumes to compensate.

Airway Resistance depends on Bronchial Smooth muscle tone

- The smooth muscles in the airway are under **autonomic control**.
- Parasympathetic cholinergic nerves cause **bronchoconstriction** → Increase airway resistance.
- Sympathetic adrenergic nerves cause **bronchodilation** → Reduce airway resistance.
- Certain chemical substances (both endogenous and exogenous) can directly alter the smooth muscle tone. Examples: Dust, smoke, or histamine (released from mast cells and basophils) can cause bronchoconstriction of the airways and induce asthmatic attacks.

Drugs, such as isoproterenol and epinephrine, stimulate β 2-adrenergic receptors in the airways, and cause bronchodilation. These agents are used to treat asthmatic attacks.

Practice Question: Multiple choice type

A 22-year-old woman is brought to the emergency department because of a 30-minute history of shortness of breath and chest discomfort. She has a 3-year history of bronchial asthma. Currently, she is not on any medication. She has history of allergy to dusts and pollen. Her temperature is 37°C (98.6°F), pulse is 105/min, blood pressure is 128/78 mm Hg, and respirations are 24/min with shallow tidal volume. Oxygen saturation in room air is 89% (Normal: 95-98%). Physical examination shows respiratory distress, wheeze, and prolonged expiration in all lung fields bilaterally. She is diagnosed with acute attack of bronchial asthma. She is treated with albuterol inhalation. Her condition improves after this treatment. Which of the following mechanisms best explains shortness of breath in this patient when she arrived?

- A. Increased lung compliance
- B. Decreased blood pressure
-  C. Increased airway resistance
- D. Release of catecholamines from adrenal medulla
- E. Sympathetic nerve stimulation

Correct Answer: C. Inhalation of some allergen must have triggered bronchoconstriction in this patient. Bronchoconstriction increases airway resistance and makes it difficult to breathe.₁₂

Dynamic Properties of the Lung: Air Flow and Airway Resistance

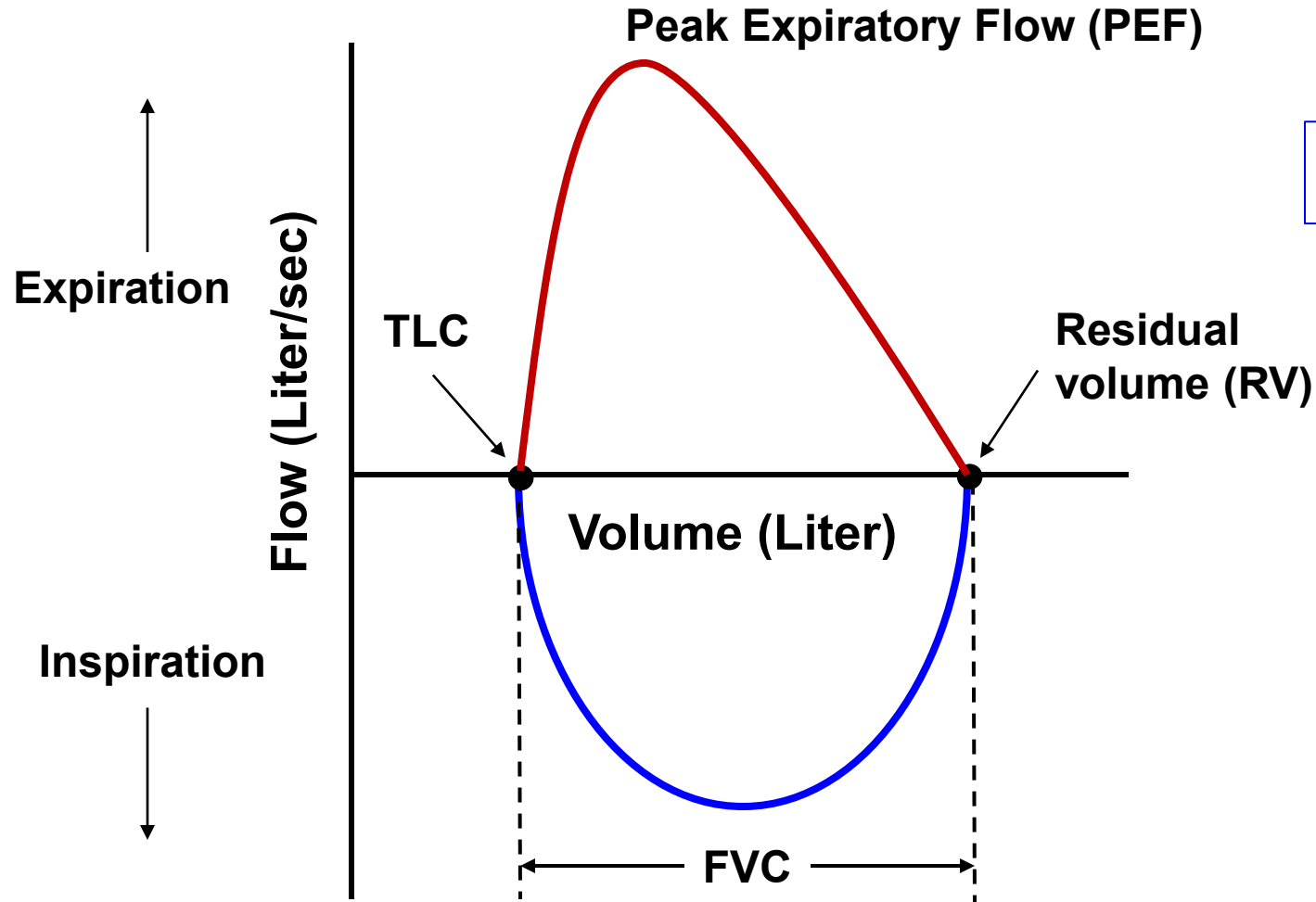
Airflow (V) is proportional to the difference between alveolar and atmospheric pressure (ΔP), but inversely proportional to airway resistance (R)

Flow = Pressure gradient between atmosphere and alveoli $\div$ Resistance

$$V = \frac{\Delta P}{R}$$

Airway resistance does not change much during normal quiet breathing. However, it increases during **forced expiration**. This alteration in airway resistance during forced expiration is a result of **airway compression**. This effect can be recorded with a **Flow-Volume curve** with forced expiration.

Flow-Volume Curve: Measured during forced breathing

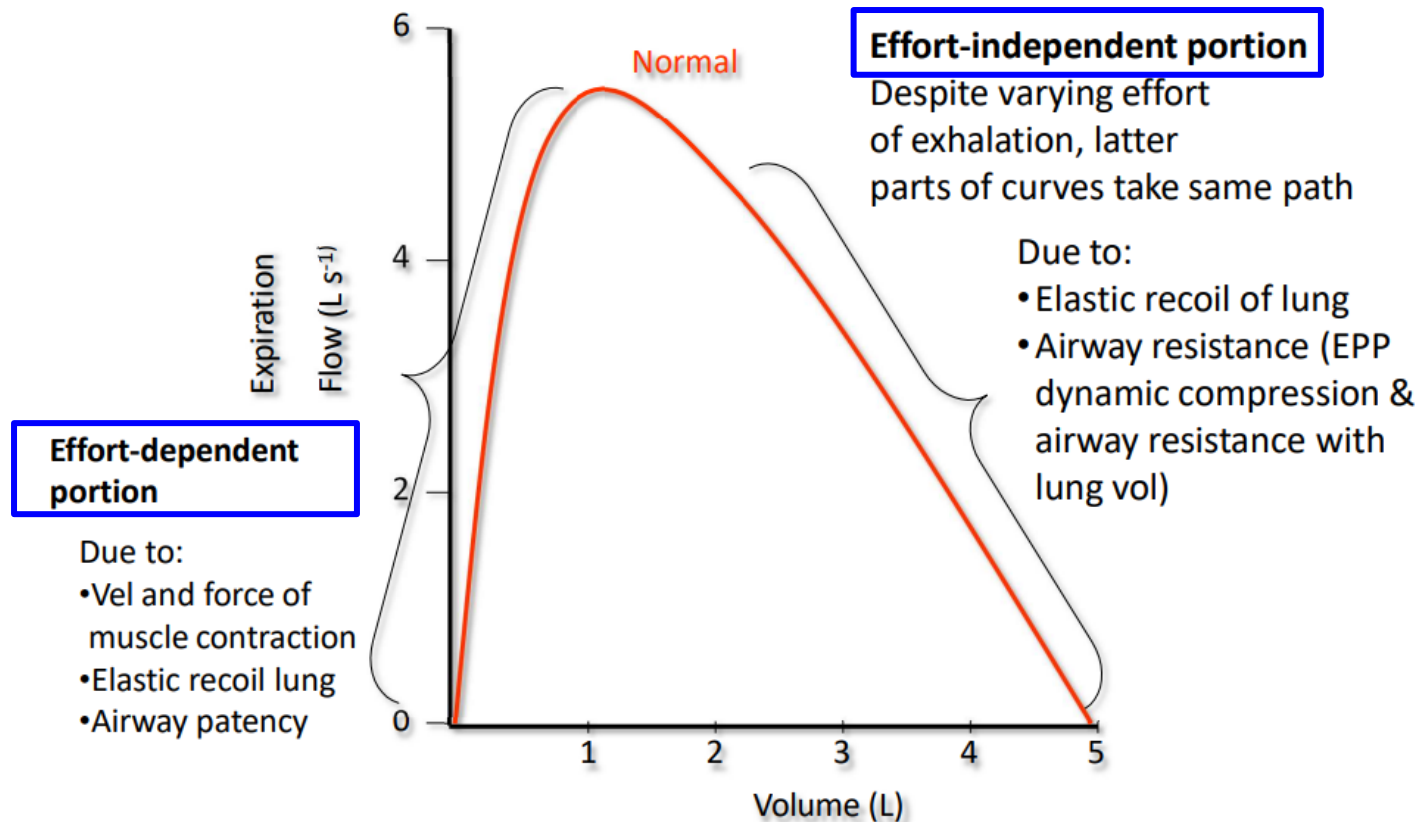


$$FVC = TLC - RV$$

Dynamic compression of the small airways is responsible for the shape of the forced expiratory flow curve

TLC= Total Lung Capacity
FVC=Forced Vital Capacity

Effort Dependent/Independent portions of the normal Flow-Volume curve



During later part of forced expiration:

↓
Intrapleural pressure becomes greater than the small airway pressure

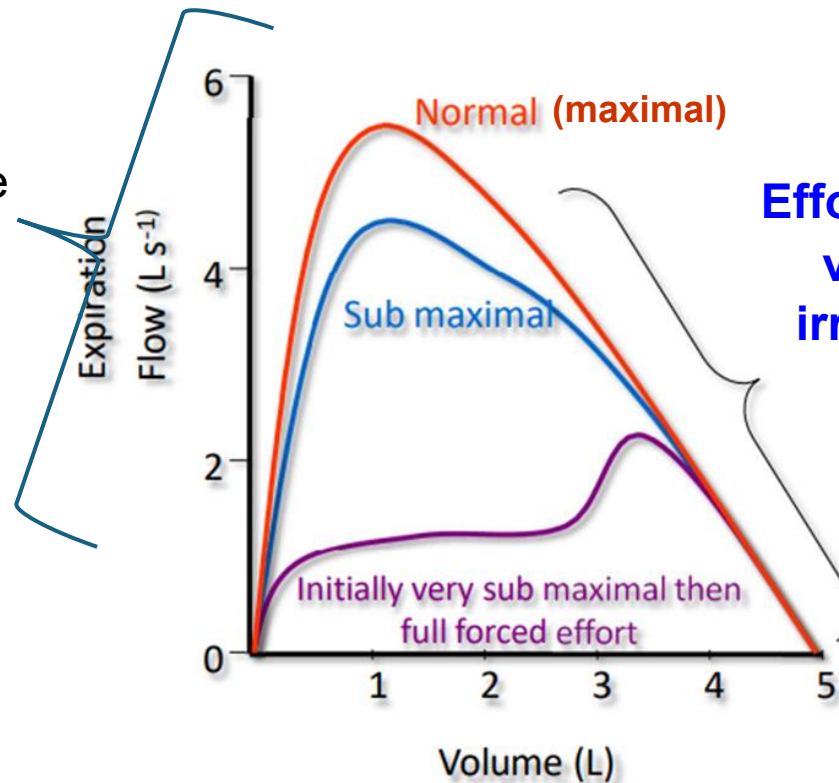
↓
Dynamic compression of the small airways

↓
Increase in airway resistance

↓
Asymmetric shape of the forced expiratory flow curve

Expiratory efforts affect only the initial Effort-dependent portion of the Flow-Volume curves

Effort-dependent portion of the Flow-volume curve varies with changing expiratory efforts



No matter how hard you try to exhale, the flow is the same towards the end of expiration

Effort-independent portion of the Flow-volume curve has the same path irrespective of the expiratory efforts

Reasons: It is due to

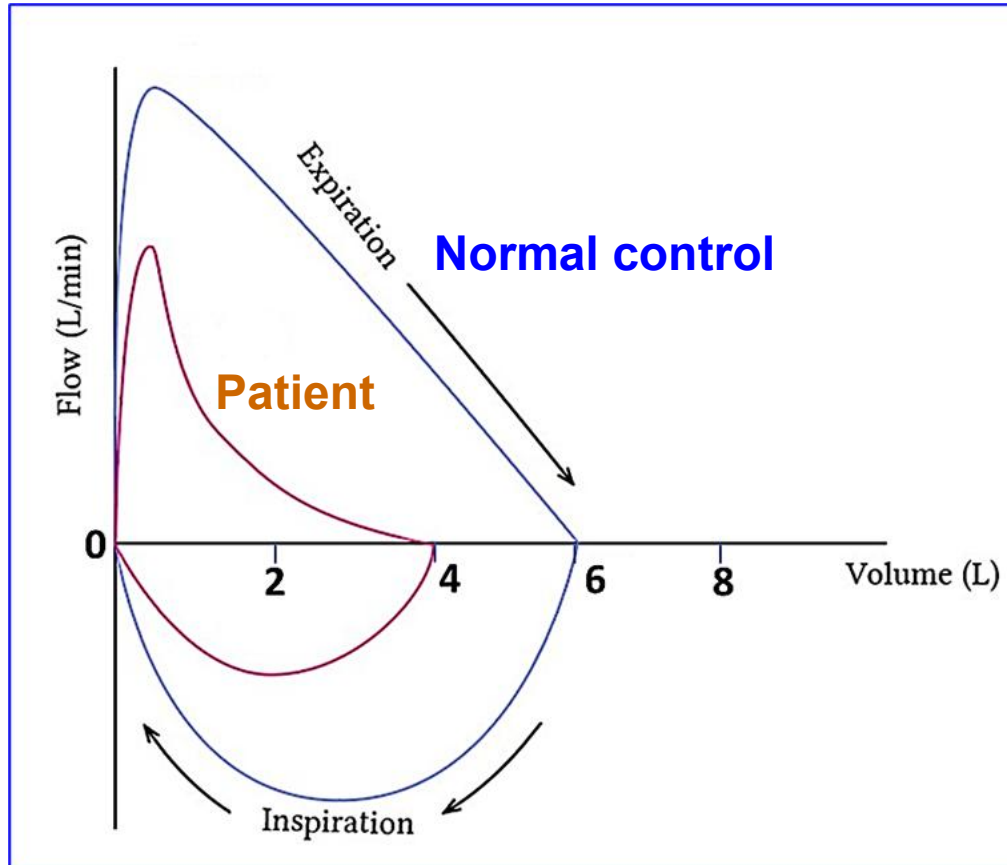
1. ↓ Elastic recoil of lung,
 2. ↓ Radial traction and
 3. ↑ Dynamic compression above the equal pressure point.
- All contribute to the increasing airway resistance in small airways

Note – time for each loop will differ. However time not displayed on this graph.

Dynamic Compression of Airways & flow-volume loop: Effort-independent portion

- During later part of forced expiration, small airways get compressed and act as a valve, rate limiting flow through the airways.
- As the lung volume declines, airway resistance increases due to a decrease in structural support (i.e., the deflated lung is less rigid because of reduced radial traction).
- As the lung volume declines, the elastic recoil of the lung reduces.
- Therefore, at low lung volumes, **greater expiratory effort does not produce greater flow**. Hence, the effort-independent portion of all flow-volume curves take the same path.

Flow-Volume curve in Obstructive Lung Diseases



During forced expiration:

↓
Intrapleural pressure becomes greater than airway pressure in longer segments of the smaller airways

↓
Significantly increased dynamic compression of the small airways

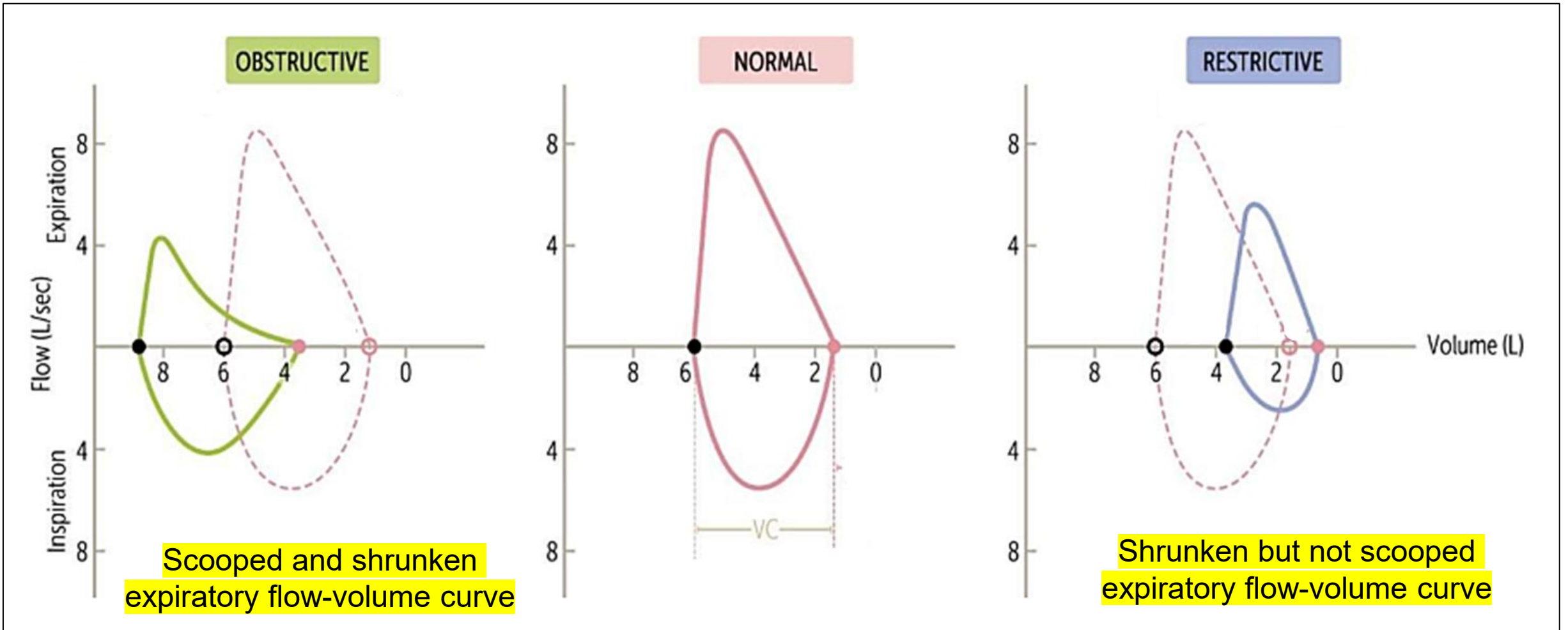
↓
Significantly increased airway resistance & prolonged expiration

↓
Scooping of the expiratory flow curve

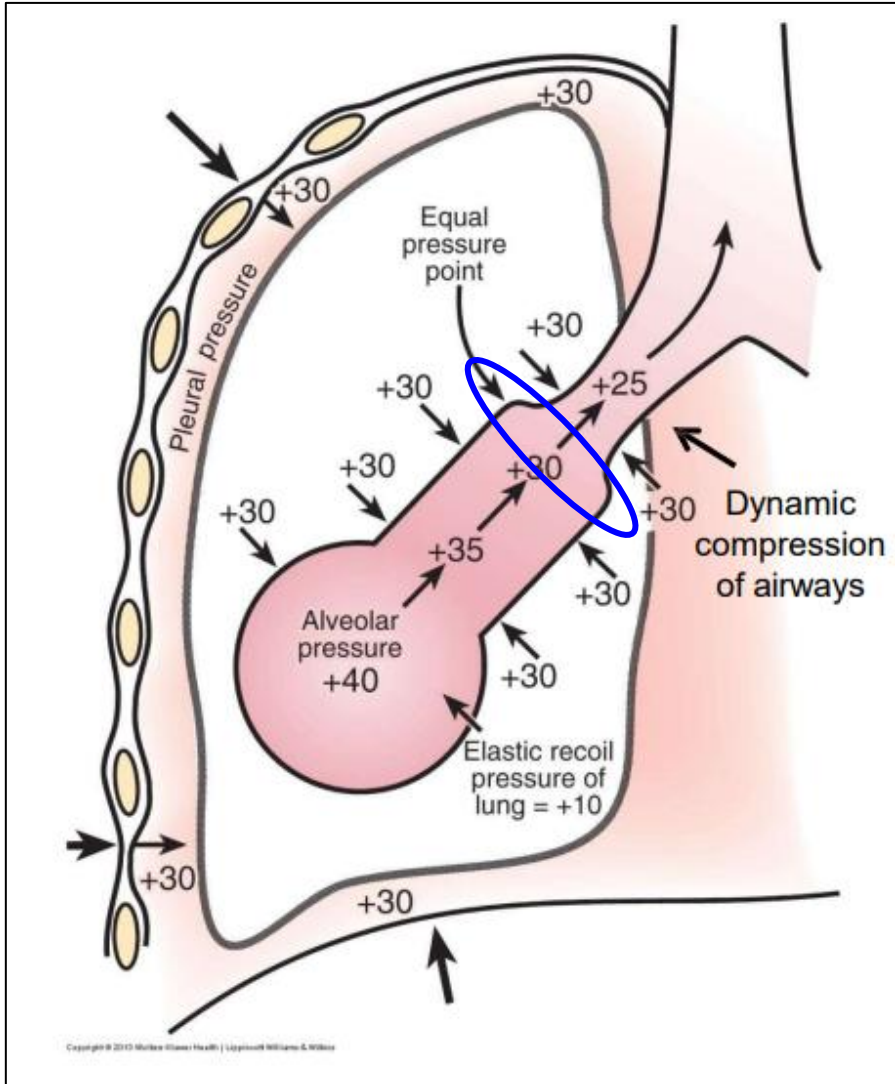
Wheezing (high-pitched whistling sound) in COPD patients is because of increased airway resistance

Flow-Volume Curves

Obstructive and Restrictive Lung Diseases



Equal Pressure Point (EPP) and Dynamic airway compression during forced expiration

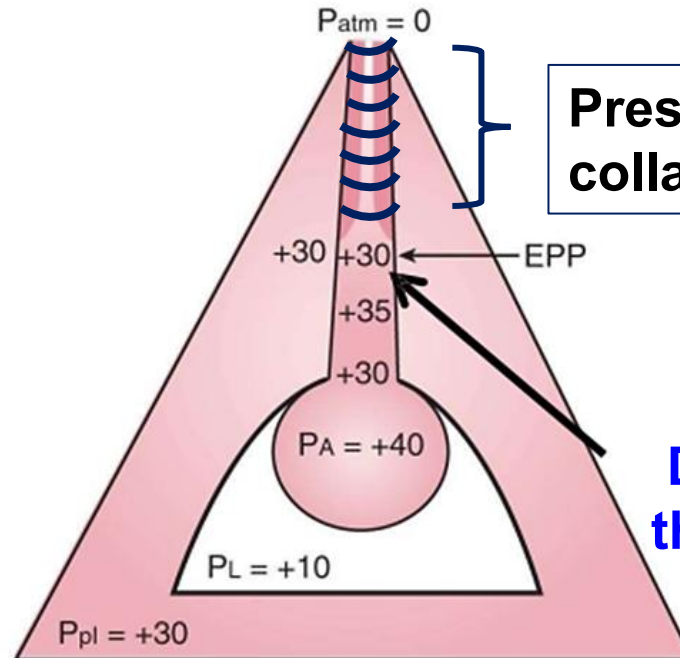


Equal Pressure Point (EPP)

- Point at which **pressure inside the small airways equals to the pressure outside** (intrapleural pressure) them during expiration
- Above this point (towards mouth), the pressure inside the airways is less than outside pressure during expiration
- Dynamic airway compression starts above the equal-pressure point (EPP) during forced expiration.

EPP in Normal Lung

EPP is located high up in the airways so that small airways (bronchioles) are well protected from dynamic compression during forced expiration.



Presence of cartilage prevents collapse of smaller bronchi

Dynamic compression starts above this point; therefore, bronchioles are well protected during forced expiration

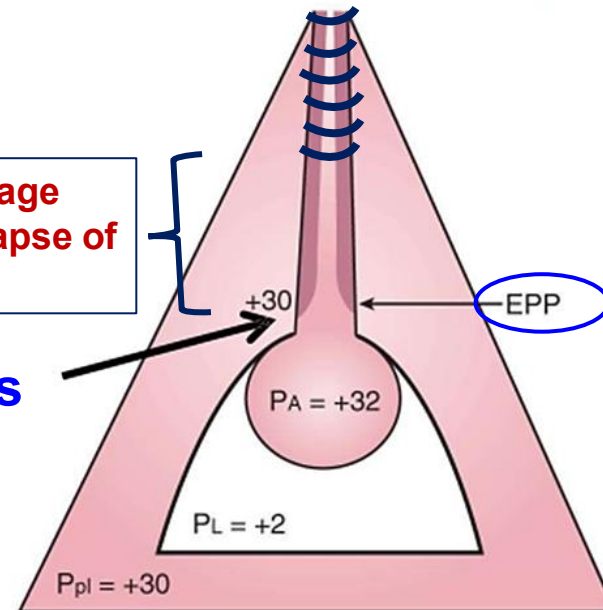
Normal lungs- EPP
Dynamic compression/restriction
typically in airways with cartilage

EPP in COPD moves downwards towards the alveoli

- Decreased elastic recoil pressure (ERP) of lung (+2 vs +10)
- Airway pressures less (+32 vs +40),
- EPP further down respiratory tree

Absence of cartilage support causes collapse of airways

Dynamic compression starts above this point and NOT protected by the cartilages



High Lung Compliance i.e. low elasticity e.g. emphysema, COPD
EPP lower down respiratory tree- dynamic compression also lower down tree.

Loss of elastic recoil (increased compliance) of the lung in COPD reduces airway pressure

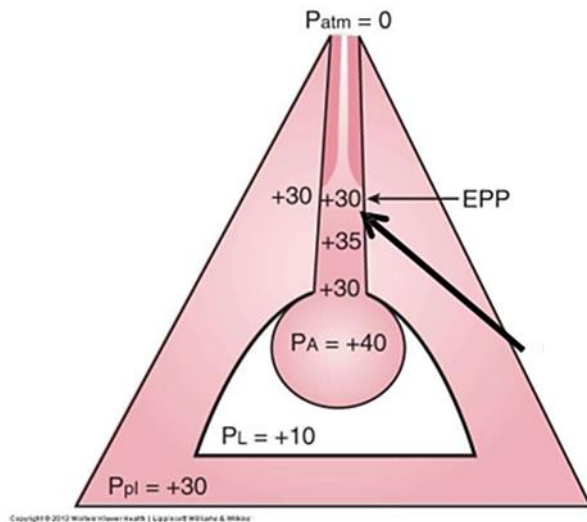
EPP moves downwards towards the alveoli

Small airways (without cartilage supports) are prone for dynamic compression during forced expiration

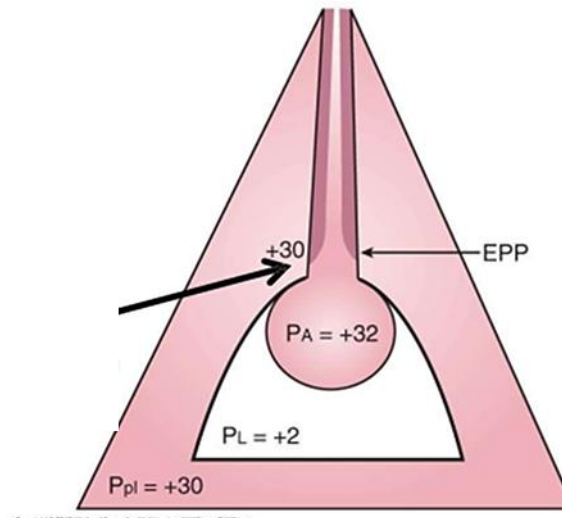
Increased airway resistance and wheeze

Practice Question: Multiple choice type

Investigators conduct a study that evaluates the effect of a new drug on the airway resistance in a population that includes both healthy individuals and patients with respiratory diseases. Investigators divide them into two groups as group1 and group2. The results of pleural and pulmonary pressures during forced expiration in these two groups are shown in the figures below. The equal pressure point (EPP) is represented by the thick black arrows in the figure. Which of the following options most likely represents these two groups?



Group 1



Group 2

Option	Group 1	Group 2
A	COPD	Fibrosis
B	Fibrosis	Healthy people
C	Emphysema	Healthy people
D	Healthy people	COPD
E	COPD	Healthy people



Correction Answer: Option D. Note that the EPP has moved more towards the alveoli in Group2 because of a reduced alveolar pressure compared to Group1. Hence Group1 must be healthy people and Group2 must be patients with COPD.

Any Questions ?

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